MD TASBEER AHMED

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MSc student in Electrical and Electronic Engineering at North South University (BSc 2024, Magna Cum Laude, CGPA 3.67) and Junior Research Fellow in the ADCS team at the Bangladesh Space Research and Remote Sensing Organization (SPARRSO), with hands-on satellite engineering experience on the NSU-AI-SAT CubeSat, as a Project Lead of the Attitude Determination and Control System (ADCS) team and a member of the communications team working on link budget analysis and antenna design. Skilled in MATLAB/Simulink modelling, Kalman-filter based sensor fusion and RF analysis; also a student researcher in swarm robotics and human-robot interaction at the NIRO Lab.

RESEARCH FOCUS

Control Systems | Swarm Robotics | CubeSat ADCS | Satellite Communications | Human-Robot Interaction | Embedded Systems

EDUCATION

North South University **Dhaka, Bangladesh**
MSc in Electrical and Electronic Engineering *Sep 2025 to present*
Thesis (Ongoing): Attitude Determination Control System of six different modes for a CubeSat

North South University **Dhaka, Bangladesh**
BSc in Electrical and Electronic Engineering, ***CGPA: 3.67 (Magna Cum Laude)*** *2020 – 2024*
Undergraduate Thesis (EEE498R): Uplink and Downlink Budget for HD Video Transmission with DVB-S2.
Capstone Project (EEE499A/B): On-campus autonomous delivery robot (Onshape 3D model, Raspberry Pi image processing, GPS localization).

Mangrove
International Advanced Level *2019*

Methodist English Medium School
IGCSE *2016*

EXPERIENCE

Junior Research Fellow (JRF), ADCS Team **Dhaka, Bangladesh**
Bangladesh Space Research and Remote Sensing Organization (SPARRSO) *2026 to present*
- Selected through the SPARRSO Research Fellowship (2026); working with the ADCS team on the design and simulation of an attitude determination and control system for an Earth-observation CubeSat.

ADCS Project Lead & Communications Team Member **Dhaka, Bangladesh**
NSU-AI-SAT CubeSat (APSCO CubeSat Competition), North South University *2024 to present*
- Contributed to the Preliminary Design Report as an undergraduate (2024); co-led the ADCS team through the Detailed Design Report.
- Built a Simulink model-in-the-loop (MIL) simulation with orbit propagation and disturbance torques (gravity gradient, solar radiation pressure, drag, magnetic).
- Implemented TRIAD and a Multiplicative Extended Kalman Filter (MEKF) for attitude and gyro-bias estimation across 6 guidance modes.
- Performed UHF/VHF and S-band link budget analysis and antenna design/simulation (CST Studio, MATLAB).

Student Researcher **Dhaka, Bangladesh**
NSU Intelligent Robotics (NIRO) Lab, North South University *2025 to present*
- Research in swarm robotics and human-robot interaction (HRI).
- Member of the NIRO Lab's Multi-UAV Drone Project team (precision agriculture), BEAR Summit 2026, BD Innovation Fair 2026.

Lab Instructor (Part-Time) **Dhaka, Bangladesh**
Department of Electrical and Computer Engineering, North South University *2024 to present*
- Conduct EEE lab sessions, teach hands-on technical work, and assess students; restructured the Control Systems lab (EEE342L) around MATLAB and Simulink.
- Teaching Assistant for Introduction to Robotics (EEE494).

Chair (2025-26), Secretary (2023-24) **Dhaka, Bangladesh**
IEEE NSU Robotics and Automation Society (RAS) Student Branch Chapter *2023 to 2026*
- Conducted workshops, seminars and webinars on robotics and automation
- Led the Chapter to two successive global awards

PROJECTS

- **NSU-AI-SAT ADCS simulation (MATLAB/Simulink):** built the disturbance-torque plant and the six-mode model-in-the-loop with TRIAD + MEKF estimation, B-dot, quaternion PID and wheel desaturation; detumbles within one orbit with under 0.7° pointing error in sunlight.
- **NSU-AI-SAT communication subsystem (MATLAB, CST Studio):** UHF/VHF and S-band link budgets and antenna selection and design using CST Studio for the Detailed Design Report.
- **NSU-AI-SAT relative-motion analysis (STK):** modelled the CubeSat's motion relative to China's Tianhe station module from TLEs for the target-tracking mode.
- **Multi-UAV Drone Project (NIRO Lab):** a three-drone coordinated spraying swarm for precision agriculture, testing whether three smaller drones can beat one large spraying drone on time and energy; BEAR Summit 2026, BD Innovation Fair 2026; selected for exhibition at CES 2027, Las Vegas.
- **Socially assistive anti-litter robots (HRI research):** built four waste-bin robots that greet passers-by and thank them for using the bin; field study at NSU gates comparing baseline, non-technical and robot interventions.

RESEARCH

- **Manuscripts in preparation**
  (1) Coordinated three-drone spraying swarm for precision agriculture (NIRO Lab);
  (2) Six-mode CubeSat attitude determination and control with TRIAD and MEKF estimation (NSU-AI-SAT).
  Both are planned for submission to a future conference.
- **Technical reports:** NSU-AI-SAT Preliminary Design Report (contributor, 2024) and Detailed Design Report (ADCS co-lead; communications), ongoing work for the APSCO CubeSat Competition.

TECHNICAL SKILLS

- **Modelling & Simulation:** MATLAB/Simulink, STK (Systems Tool Kit), KiCad, CST Studio Suite, Cadence, ModelSim, Multisim, Logisim.
- **Estimation & Control:** B-dot, quaternion PID, Link Budget Analysis, Sensor fusion (TRIAD, MEKF)
- **Programming & Hardware:** Python, C, Arduino and microcontrollers (Raspberry Pi Pico, Raspberry Pi).
- **Languages:** Bangla (native), English

AWARDS & RECOGNITION

- Multi-UAV Drone Project (NIRO Lab) selected for exhibition at CES 2027, Las Vegas.
- First Runner-up, Undergraduate Thesis Paper Contest, IEEE Student Led Con 2023 (for the DVB-S2 thesis).
- IEEE Robotics and Automation Society Student Branch Chapter of the Year Award 2025
  - Award received at the **International Conference on Robotics and Automation (ICRA 2025)**
- IEEE Robotics and Automation Society Student Branch Chapter of the Year Award 2026
- Best Event Award, RAS Day 2025 (IEEE NSU RAS Student Branch Chapter).

REFERENCES

Dr. Shahnewaz Siddique, Associate Professor, Department of Electrical and Computer Engineering, North South University
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Dr. Lamia Iftekhar, Department of Electrical and Computer Engineering, North South University
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